

Module 1

1. Implementing the Singleton Pattern

Code:

```
class Logger {

    // Single instance
    private static Logger instance;

    // Private constructor
    private Logger() {
        System.out.println("Logger instance created.");
    }

    // Method to return the single instance
    public static Logger getInstance() {
        if (instance == null) {
            instance = new Logger();
        }
        return instance;
    }

    // Log method
    public void log(String message) {
        System.out.println("Log: " + message);
    }
}

public class SingletonPatternExample {
```

```
public static void main(String[] args) {

    Logger logger1 = Logger.getInstance();
    Logger logger2 = Logger.getInstance();

    logger1.log("First message");
    logger2.log("Second message");

    // Check if both objects are the same
    if (logger1 == logger2) {
        System.out.println("Only one Logger instance exists.");
    } else {
        System.out.println("More than one Logger instance exists.");
    }
}
}
```

Output:

```
[Running] cd "d:\Documents\Internships and Hackathons\Cognizant
Java Digital Nurture\Coding\Module 1\" && javac
SingletonPatternExample.java && java SingletonPatternExample
Logger instance created.
Log: First message
Log: Second message
Only one Logger instance exists.

[Done] exited with code=0 in 1.793 seconds
```

2. Implementing the Factory Method Pattern

Code:

```
// Document Interface
interface Document {
    void open();
}

// Word Document
class WordDocument implements Document {
    public void open() {
        System.out.println("Word Document Created.");
    }
}

// PDF Document
class PdfDocument implements Document {
    public void open() {
        System.out.println("PDF Document Created.");
    }
}

// Excel Document
class ExcelDocument implements Document {
    public void open() {
        System.out.println("Excel Document Created.");
    }
}

// Abstract Factory
```

```
abstract class DocumentFactory {  
    abstract Document createDocument();  
}  
  
// Word Factory  
class WordFactory extends DocumentFactory {  
    public Document createDocument() {  
        return new WordDocument();  
    }  
}  
  
// PDF Factory  
class PdfFactory extends DocumentFactory {  
    public Document createDocument() {  
        return new PdfDocument();  
    }  
}  
  
// Excel Factory  
class ExcelFactory extends DocumentFactory {  
    public Document createDocument() {  
        return new ExcelDocument();  
    }  
}  
  
// Main Class  
public class FactoryMethodPatternExample {  
  
    public static void main(String[] args) {
```

```
DocumentFactory wordFactory = new WordFactory();  
Document word = wordFactory.createDocument();  
word.open();
```

```
DocumentFactory pdfFactory = new PdfFactory();  
Document pdf = pdfFactory.createDocument();  
pdf.open();
```

```
DocumentFactory excelFactory = new ExcelFactory();  
Document excel = excelFactory.createDocument();  
excel.open();
```

```
}
```

```
}
```

Output:

```
[Running] cd "d:\Documents\Internships and Hackathons\Cognizant  
Java Digital Nurture\Coding\Module 1\" && javac  
FactoryMethodPatternExample.java && java  
FactoryMethodPatternExample  
Word Document Created.  
PDF Document Created.  
Excel Document Created.  
  
[Done] exited with code=0 in 1.526 seconds
```